THE BIG TASKS BEFORE ISRO

With Shukla going up in space, what India gains is the good exposure and experience needed for human interface processes, which is important, but only one part of the entire programme. India not only has to build and validate a suitable launch vehicle but also orbital and crew modules. This is in addition to training astronauts and ensuring their safety at launch, in space and on their return to earth. "It's a whole new and challenging game for us, but our scientists and collaborating institutions are up to it," says Narayanan.

Among the first things ISRO did was to build Launch Vehicle Mark3, or LVM3, which would be capable of lifting heavy payloads of 10 tonnes to low earth orbit. To make these launch vehicles human-rated, numerous test firings will be needed. The redundancies, too, have been trebled so that if one system fails, there are enough back-ups to prevent a catastrophe. Narayanan says, "When there is human life involved, we have to be thrice as careful. We have to score 100 out of 100 when it comes to safety protocols." In fact, at the Ax-4 mission involving Shukla, it was Narayanan who refused to give his clearance for launch after mission control reported a slight glitch. His insistence saw the Axiom technicians discover a hairline crack in one of the engines that could have proved disastrous on flight. Only after it was rectified did Narayanan give his consent.

Back home, on its own projects, what ISRO had to do from scratch was build a crew module that can carry three or four astronauts with adequate Environmental Control and Life Support Systems (ECLSS). These maintain comfortable room temperatures, the right cabin pressure and adequate oxygen levels, apart from food, living quarters and waste disposal. Former ISRO chairman S. Somanath points out, "These are essentially 'do it yourself' or DIY technologies. Initially we thought we could just buy these and then later develop them. But they didn't materialise because they came at a high cost and were unable to meet our schedules." ISRO also had to develop the technologies for the re-entry of the crew vehicle, then the soft landing at sea using parachutes, and recovery by the armed forces. It has also tested the crew escape system where the crew module is ejected at a safe distance in case of rocket failure on the launchpad or during lift-off.

The other systems ISRO has to perfect are communications, navigation and a high degree of automation. As important is the selection and training of astronauts. The first four astronauts for Gaganyaan, including Shukla who went up in Axiom, are all Indian Air Force pilots selected after a range of tough tests. Besides developing an astronaut training centre in Bengaluru that currently focuses on physical and mental fitness, the astronauts have been sent to the cosmodrome in Moscow for rigorous tests, including a sub-orbital flight to experience zero gravity briefly. The test is called vomit-comet because, invariably, the trainees throw up on their first attempt.

INDIA'S BIG LEAP

Two missions, each a stepping stone for the next, symbolise India's grand bid to be a spacefaring nation



GAGANYAAN

AIM: Launch a crew of three Indian astronauts into space and bring them back safely

TARGET: 2027

PRECURSORS

- Three uncrewed missions, beginning by the end of 2025
- The first will have Vyommitra, a female humanoid robot, on board

LESSONS FROM SHUKLA'S TRIP

Rakesh Sharma recalls that no amount of training on the ground helps you prepare for all that happens in orbit, including withstanding G-force or motion sickness. Shukla himself revealed that he found "drinking water, walking, sleeping a big challenge—you can sleep on the roof, or the walls and on the ground". Aboard the ISS, astronauts sleep in compartments the size of a phone booth and equipped with sleeping bags that are secured to the walls of the spacecraft, so that they don't float away. They cannot use showers to bathe as water doesn't flow in microgravity. Instead, as Sharma did 40 years ago, they use wet wipes, rinseless shampoo and liquid soap. Shukla seemed to have dispensed with shaving as well, sporting a thick stubble within days of arrival. Going to the toilet, as Sharma recalls, "is a well-practised art because even your bodily effluents are weightless just as you are. So, you have to make sure that they are captured effectively. It does take some practice." Space food, though, has evolved and moved beyond the food tubes that Sharma snacked on. Shukla can choose between vacuum-packed, freeze-dried, dehydrated or canned items. He has also carried some *gajar ka halwa*, *moong dal halwa* and